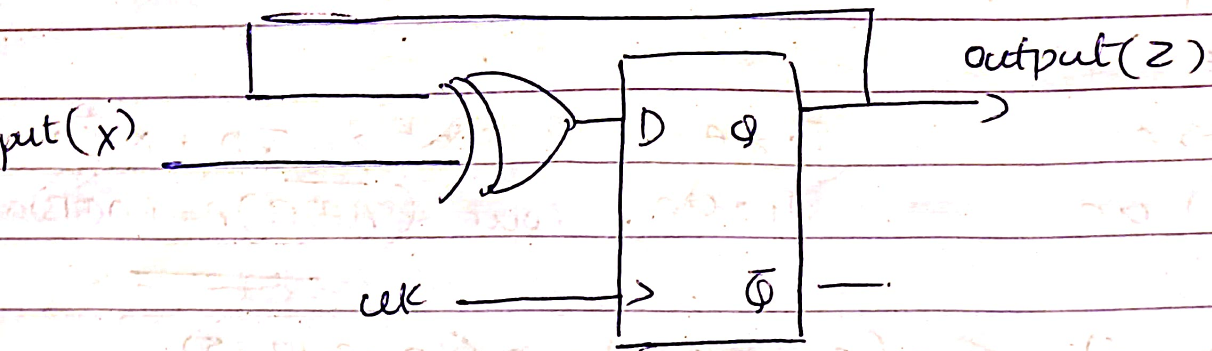
Assignment - Day 4 - Sequential

1) Explain the behaviour of the given circuit



XOR gate inputs are

External input = x

Feedback input = Q

$D = x \oplus Q$

$Q_{next} = D$

$Q_{next} = x \oplus Q$

$Z_{next} = x \oplus Z$

State Table

Present state (Q)	Input (x)	Q_{next}
0	0	0
0	1	1
1	0	1
1	1	0

case 1 : $x = 0$ $\Rightarrow Q_{next} = Q$

case 2 : $x = 1$ $\Rightarrow Q_{next} = \bar{Q}$

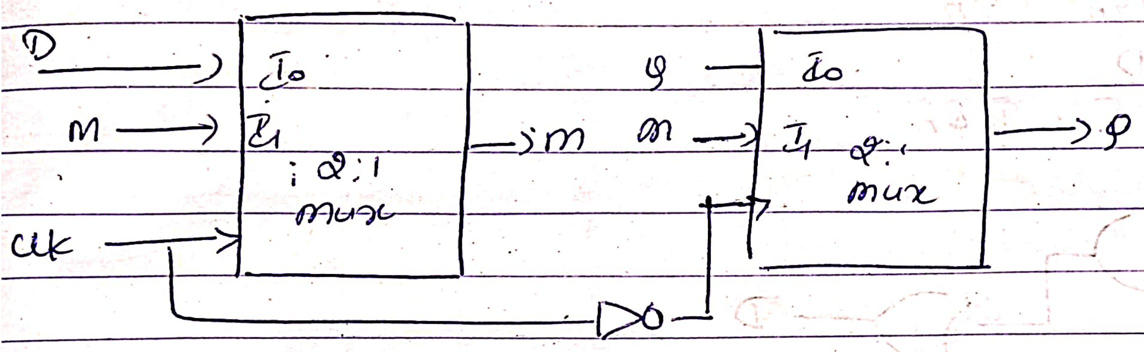
The circuit behaves like a T flip flop

2. Design a posedge triggered D FF using 2:1 Muxes only

Master Latch $\Rightarrow M = \overline{clk} D + clk M$

Slave latch $\Rightarrow Q = clk\, M + \overline{clk}\, Q$

clk = 0		clk = 1	
Mux1 input	Connection	Mux 2 Input	connection
I_0	D	I_0	Q (feedback)
I_1	M (feedback)	I_1	M
Select	clk	Select	clk
Output	M	Output	Q



3. Design a negedge Triggered T-FF using 2:1 muxes only

Characteristic equation of TFF is $Q_{next} = T \oplus Q$

For 2:1 Mux

select line - T

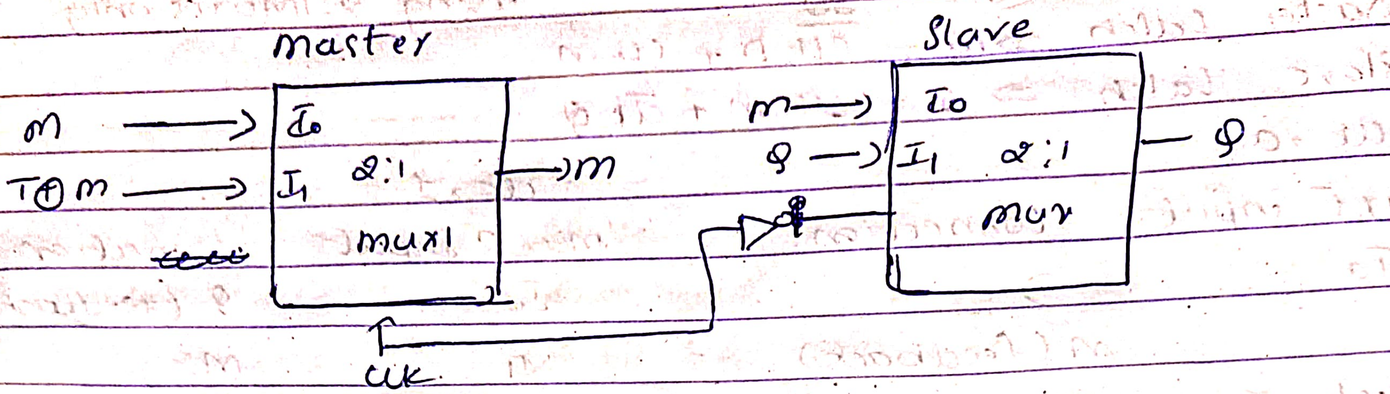
$I_0 = Q$ $\qquad$ $M = \overline{clk} M + clk\,(T \oplus M)$

$I_1 = \overline{Q}$ $\qquad$ $Q = clk\, Q + \overline{clk}\,(M)$

$y = \overline{T} Q + T\overline{Q}$

$y = T \oplus Q$

Master latch clk = 1		Slave latch, clk = 0	
Input	Connection	Input	Connection
I_0	M (feedback)	I_0	M
I_1	$T \oplus M$	I_1	Q (feedback)
select	clk	select	clk
Output	M	Output	Q



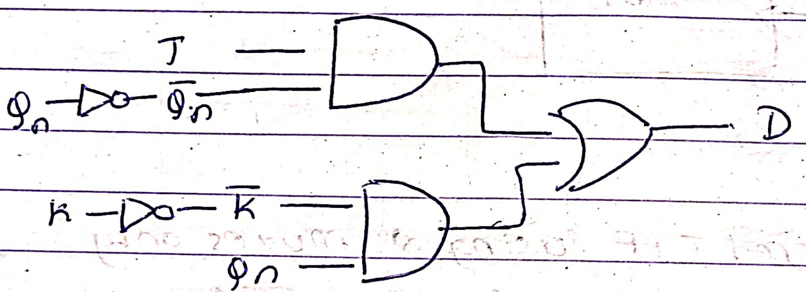
4) convert by adding external gates:

a) A D flip-flop to a JK flip-flop

$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$

$Q_{n+1} = D$

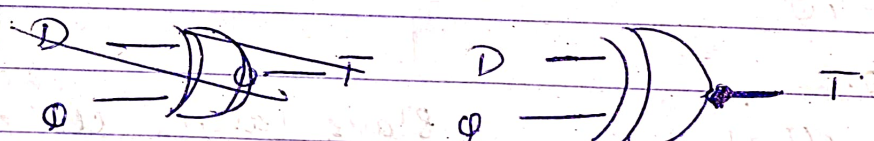
$D = J\bar{Q}_n + \bar{K}Q_n$



b) A T Flip-Flop to a D flip flop

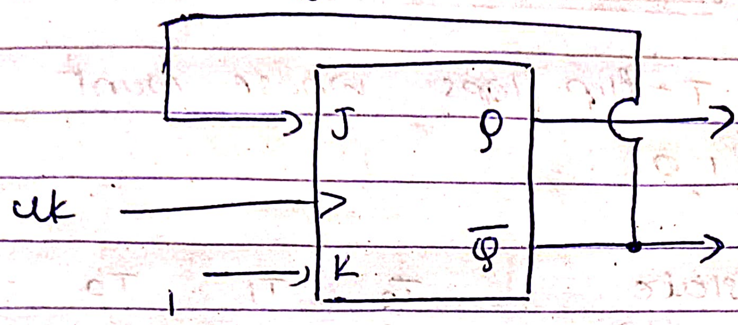
$Q_{n+1} = T \oplus Q$ $\quad Q_{n+1} = D \quad \Rightarrow D = T \oplus Q$

$T = D \oplus Q$



5. In a positive edge Triggered JK flip flop, we have $J = Q'$ and $K = 1$ as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for

those 6 clock cycles



$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$

$K = 1 \Rightarrow \bar{K} = 0$ $J = \bar{Q}$

$Q_{n+1} = \bar{Q} \cdot \bar{Q} + 0$

$Q_{n+1} = \bar{Q}$ (Toggles on every clock pulse)

clock pulse	present Q	J, Q'	K	Next Q
0	0	1	1	1
1	1	0	1	0
2	0	1	1	1
3	1	0	1	0
4	0	1	1	1
5	1	0	1	0
6	0	1	1	1

Sequence is $0 \to 1 \to 0 \to 1 \to 0 \to 1 \to 0$